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A place to talk about Modular Synthesizers. And Cats.

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303 circuit modelling

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Re: 303 circuit modelling

by [guest](#) » Mon Mar 03, 2025 10:47 pm

oof, well this takes the wind out of my sails a bit:

127 posts

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Super Deluxe WigglerPosts: 13314
Joined: Mon Aug 19, 2013 11:49 am[openmusiclabs.com](#)

Re: 303 circuit modelling

by [toneburst](#) » Tue Mar 04, 2025 9:54 am

I know that feeling, when you've been working on a project for ages, then someone else comes out with a similar thing. Don't give up though!

[toneburst](#)
Super Deluxe WigglerPosts: 1042
Joined: Fri May 02, 2014 2:41 pm
Location: London

Re: 303 circuit modelling

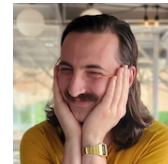
by [Luap](#) » Tue Mar 04, 2025 10:16 amI was just about to say "Happy 303 day!" then I realised it's the 4th... Oh well. It's the thought that counts, right??
Keep up the good work Guest![Luap](#)
Super Deluxe WigglerPosts: 1262
Joined: Fri Nov 30, 2012 6:41 pm
Location: Cornwall, UK

Re: 303 circuit modelling

by [stephven](#) » Tue Mar 04, 2025 10:19 am

well i can't read japanese and there doesn't seem to be any info on the device in the video otherwise, so i'll def stay subbed to this thread.

I appreciate the documentation and a look into VA synth design.

[stephven](#)
Veteran WigglerPosts: 709
Joined: Sun Jun 07, 2020 10:41 am
Location: Indianapolis, USA
Contact: [stephven](#)

(they/them)

i'm making Eurorack and 5U stuff: <https://www.gauche-electrosound.net>new stuff: tiny pieces
harsh noise vaporwall
12 Sep 2025
[tiny pieces](#)

Re: 303 circuit modelling

by [bmot](#) » Tue Mar 04, 2025 10:55 am

I'm impressed, I found it very hard to tell the difference between the HYP3 Core Engine and the real 303.

[stephven](#) wrote:

Tue Mar 04, 2025 10:19 am

well i can't read japanese and there doesn't seem to be any info on the device in the video otherwise, so i'll def stay subbed to this thread.

I appreciate the documentation and a look into VA synth design.

I translated the info under the video and it say:

"This is a record of a comparison test conducted in June 2024 between an analog modeling sound source called HYP3 Core Engine (tentative name) and Roland TB-303. HYP3 Core Engine (tentative name) is a modeling sound source used in hardware synthesizers under development at SYNTHERNET, such as HYP3 and G3N3. The MICRO HYP3 used in the video is experimental hardware. It runs on an overclocked STM32F411 (Cortex-M4, 120MHz). The display in the upper left uses a spectrograph called as_SonoGx2. The horizontal axis is time and the vertical axis is frequency. The resonant peaks are visible as bright lines, so you can observe the envelope shape of the cutoff frequency. When orange (HYP3) and blue (TB-303) overlap, they are displayed as white. In Test 1, we compare the state in which the envelope rises in response to an increase in tempo and a series of accents. Since the oscillator is free running, there is a phase difference, but the envelope itself is almost the same. In test 2, we compare the changes when changing the parameters. The HYP3 Core Engine matches the potentiometer curve and parameter range to the TB-303. This can also be adjusted to suit the individual unit. In test 3, we compare the results when the same external distortion is applied. Even when distorted, you can get a very close feeling."

[Modular Grid](#)
[bmot YouTube channel](#)[bmot](#)
Super Deluxe WigglerPosts: 2470
Joined: Sun Jan 22, 2012 8:12 am
Location: Germany
Contact: [bmot](#)

[Flysian Park - 80s synth pop \(Bandcamp\)](#)
[Sootify](#)

Re: 303 circuit modelling

by [guest](#) » Tue Mar 04, 2025 12:48 pm

i appreciate all the comments, and will definitely keep working on it, but maybe at a slower pace. the size and layout they have (including the little round buttons) was exactly what i was envisioning, a little pocket-x0x. i had another idea for integrating it into a game controller, so joysticks for controlling reso and cutoff, so maybe ill go that direction instead.

I have a workaround for the reso issue, but i dont like it, so im going to try redoing my naive integrators with trapezoidal integrators, although i expect that to be worse as the delay is longer.

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Re: 303 circuit modelling

by [toneburst](#) » Wed Mar 05, 2025 5:45 am

Yes, do keep up the great work! I like the game controller idea. Maybe with some algorithmic pattern-generator integrated too. That would be super fun!

Re: 303 circuit modelling

by [toneburst](#) » Wed Mar 12, 2025 5:39 am

Hi @guest as requested, posting my SuperCollider-based 303 emulator silly questions here.
Thanks so much for agreeing to help!

My confusion is mainly around the filter and filter envelope, the ranges and responses of the various controls and the way they interact.

- What's the approximate range of the filter cutoff in Hz, with Env Mod set to its minimum value?
- Is the response of the filter cutoff control linear, exponential, or does it have some other curve?
- Does the filter envelope apply a positive or bipolar offset to the cutoff frequency (the former I think, but maybe it's not that simple)?
- Does the offset applied by the filter envelope remain fixed across the sweep of the cutoff frequency control?
- If yes to the above, what is the approximate maximum non-accented offset applied by the filter envelope to the base cutoff frequency?
- If no, how does the filter envelope scale when base cutoff freq is swept?
- Is the Env Mod control linear in response?
- Does base cutoff freq track oscillator pitch at all?

I think this is probably enough questions for now!

Re: 303 circuit modelling

by [guest](#) » Wed Mar 12, 2025 12:26 pm

1. cutoff frequency is controlled by an exponential pot taper, but goes to a linear control input, so its mostly exponential. the exact tuning of the 303 is set by TM3, and i cant get mine to be in spec, but is pretty close to spec (service manual settings). for cutoff = 0, and envmod = 0, it goes from 210Hz to 1.7kHz. resonant frequency is measured at resonance = full up. the resonant frequency changes with resonance setting.

2. the cutoff frequency is essentially an exponential with all controls in the exponent: $F = e^{(envmod + cutoff + accent)}$. the envmod signal is bipolar, and the envmod pot controls the cutoff frequency, even when there is no envelope present. this is because "no envelope" is a value of 0V DC on the mod input, and it is bipolar with respect to a 3.2V reference, so as you turn the envmod pot up, you are adding less of the 0V (or -3.2V if you want to think of it that way). so with envmod full up, the above numbers are 77Hz and 620Hz. when there is an envelope present, the peak is 10V, so for envmod all the way down you get 462Hz and 4.1kHz, and for envmod full up you get 4.167kHz and 36kHz. the pot for envmod is expo, and it controls the current exponentially, which makes it a double expo, which is wierd and i dont know why i didnt realize that until now. the acutal envelope just controls things expo wise, but the pot makes the manual control a double expo.

3. base cutoff frequency does not track pitch.

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Re: 303 circuit modelling

by [toneburst](#) » Wed Mar 12, 2025 2:51 pm

Cool, thank you!! I will digest that. I. Sure I'll have more questions, but that's super helpful.

I may not get round to testing stuff out for a little while now, sadly, as my laptop screen (I'm hoping just the backlight in fact) has died (typing this on my iPad, but no SuperCollider here).

Re: 303 circuit modelling

by [guest](#) » Tue Apr 08, 2025 5:18 pm

oh my god, i thought i was never going to work on this project again. i spent weeks trying to figure out the resonance problem, and then that other group released their version, which was quite good, and it was all quite demoralizing. so i stepped away for the past month and worked on other things (i have some fun RP2350 stuff in the works!). i did a bit of reading up on filter issues, but mostly thought it was beyond my capabilities to get this thing to work. but, today i returned to it, set everything back up, and tried an idea i got from one of the papers i read, and it worked! literally 5 minutes of work. i changed one line of code and now ive got the resonance i want. i can not tell you how happy i am. this thing was really getting me down. at any rate, now i need to dig in and try to understand why the change i made worked. it makes no sense to me.

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Re: 303 circuit modelling

by [batchas](#) » Wed Apr 09, 2025 2:33 am

[stephen](#) wrote:

well i can't read japanese and there doesn't seem to be any info on the device in the video otherwise, so i'll def stay subbed to this thread.

There's an online tool that I really appreciate and find pretty good for translations:

<https://www.deepl.com/translator>

You type or copy/paste the text to translate, japanese text for instance and it will translate for you in the language of your choice.

Comment on YouTube by @hipster1966 about The Orb - Live at Manchester Academy in 1992:

"I took some acid & spent the night looking in the wrong direction - I met a friend who pointed me towards the stage"

My albums on Bandcamp: <https://batchas.bandcamp.com/>

My modular journey on YouTube: <http://www.youtube.com/batchas>

Re: 303 circuit modelling

by [batchas](#) » Wed Apr 09, 2025 2:41 am

[guest](#) wrote:

oh my god, i thought i was never going to work on this project again. i spent weeks trying to figure out the resonance problem, and then that other group released their version, which was quite good, and it was all quite demoralizing. so i stepped away for the past month and worked on other things (i have some fun RP2350 stuff in the works!). i did a bit of reading up on filter issues, but mostly thought it was beyond my capabilities to get this thing to work. but, today i returned to it, set everything back up, and tried an idea i got from one of the papers i read, and it worked! literally 5 minutes of work. i changed one line of code and now ive got the resonance i want. i can not tell you how happy i am. this thing was really getting me down. at any rate, now i need to dig in and try to understand why the change i made worked. it makes no sense to me.



[guest](#)
Super Deluxe Wiggler

Posts: 13314

Joined: Mon Aug 19, 2013 11:49 am

[toneburst](#)

Super Deluxe Wiggler

Posts: 1042

Joined: Fri May 02, 2014 2:41 pm

Location: London

[toneburst](#)

Super Deluxe Wiggler

Posts: 1042

Joined: Fri May 02, 2014 2:41 pm

Location: London



[guest](#)
Super Deluxe Wiggler

Posts: 13314

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[toneburst](#)

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Posts: 1042

Joined: Fri May 02, 2014 2:41 pm

Location: London



[guest](#)
Super Deluxe Wiggler

Posts: 13314

Joined: Mon Aug 19, 2013 11:49 am



[batchas](#)
Super Deluxe Wiggler

Posts: 6349

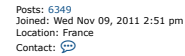
Joined: Wed Nov 09, 2011 2:51 pm

Location: France

Contact: [✉](#)



[batchas](#)
Super Deluxe Wiggler



Comment on youTube by @hipster1966 about The Orb - Live at Manchester Academy in 1992:
 "I took some acid & spent the night looking in the wrong direction - I met a friend who pointed me towards the stage"

Posts: 91
Joined: Tue Aug 21, 2018 10:24 pm

that sounds like its running too fast, everything is pitched up. i should double check that i have the latest code up. the ADC inputs are 0-1023, linear.

i dont know about the array thing, i usually do lookup tables as

```
int myarray[1024] = { put lookup table here, csv file};
```

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Re: 303 circuit modelling

■ by [sphet](#) » Tue Apr 22, 2025 11:03 am

Hey guest,

I am not sure why it is pitched up or faster. I read from your earlier post that it is running at 50kss, so I have a 50000kps buffer being filled with one value from analogWrite() each sample. I copy the same value to the left and right sides. Am I wrong in the sample rate your ISR is running at?

I notice in the code that you are subtracting 0x500 from the output before writing it - what is this for? If the value were unsigned i could imagine in is making it quieter, but given that it is signed doesn't it just shift the entire generated waveform off of zero?

Re: 303 circuit modelling

■ by [guest](#) » Tue Apr 22, 2025 11:29 am

oh yes, maybe that is it. im outputting on the 12b DAC, which is unsigned. so maybe you can get rid of that offset and scaling and see what happens.

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Re: 303 circuit modelling

■ by [toneburst](#) » Wed Apr 23, 2025 4:50 am

A little update on my 303 emulation progress for my Norns project: I gave up on my own SuperCollider emulation attempt in the end, and opted for writing a SuperCollider plugin wrapper around the DSP code from <https://github.com/RobinSchmidt/Open303>.

I have very little experience with C++, so that turned out to be a major project! I feel I learnt a lot though, and it's now working pretty much as intended, and sounding... very like a 303.

I hacked in a morphing (low>band>high-pass) filter, and may add other features (sub-oscillator, maybe noise, modulation of filter cutoff from noise/oscillator).

Here's a little example showing the basic 303 sound, plus filter morphing:
<https://soundcloud.com/toneburst/bline-...-ne-demo-03>

I need to work on level-management. I've since added a limiter, but I'll need to add in a compressor too, to even levels out a bit, especially with low resonance, and with band and high-pass filters.

Thanks very much @guest for your advice. I'm continuing to watch this project with interest, as I think emulating the 303 on low-cost/power microcontrollers is still very much a worthwhile project. Keep up the great work!

Re: 303 circuit modelling

■ by [guest](#) » Wed Apr 23, 2025 12:04 pm

that sounds really good! and thanks for the link to the other project, i hadnt seen that yet.

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Re: 303 circuit modelling

■ by [toneburst](#) » Thu Apr 24, 2025 5:29 am

“ guest wrote: “

Wed Apr 23, 2025 12:04 pm

that sounds really good! and thanks for the link to the other project, i hadnt seen that yet.

No problem! I take no credit for the DSP work. It might be worth looking at, but my impression is that it's very much written with desktop computers, rather than MCU use in mind, and probably isn't very well optimised (for a start, every variable is double-precision floating-point).

It does sound pretty good though. There's a long thread about the project on KVR. It sounds like it was quite a community effort.

<https://www.kvraudio.com/forum/viewtopic.php?t=262829>

Incidentally, my morphing filter addition is a very naive implementation. The filter class already has an option to switch to a variety of different modes. I simply added a second instance of the filter and did some mode-switching and crossfading of the two filters audio outputs. I could probably have done it by modifying a single filter, and interpolating coefficients and pole-mixing values, but I suspect that would probably be as expensive as my brute-force method, ultimately.

Re: 303 circuit modelling

■ by [guest](#) » Thu Jun 05, 2025 12:49 am

its been a while, things got busy for a bit, but im back at it. i have to make wavetables, and i need to decide how many tables to make. the saw is a pure saw, so that decision is just about cutoff frequency limitations, but the square is more complicated. i may need to make a lookup table per note. here are overlays of the square across 3 octaves, from the lowest note to the highest note the 303 can do (it can actually go one octave higher with pattern shifting, but the changes become less and less the higher you go).



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



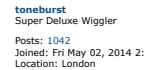
[sphet](#)
Common Wiggler

Posts: 91
Joined: Tue Aug 21, 2018 10:24 pm



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



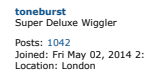
[toneburst](#)
Super Deluxe Wiggler

Posts: 1042
Joined: Fri May 02, 2014 2:41 pm
Location: London



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



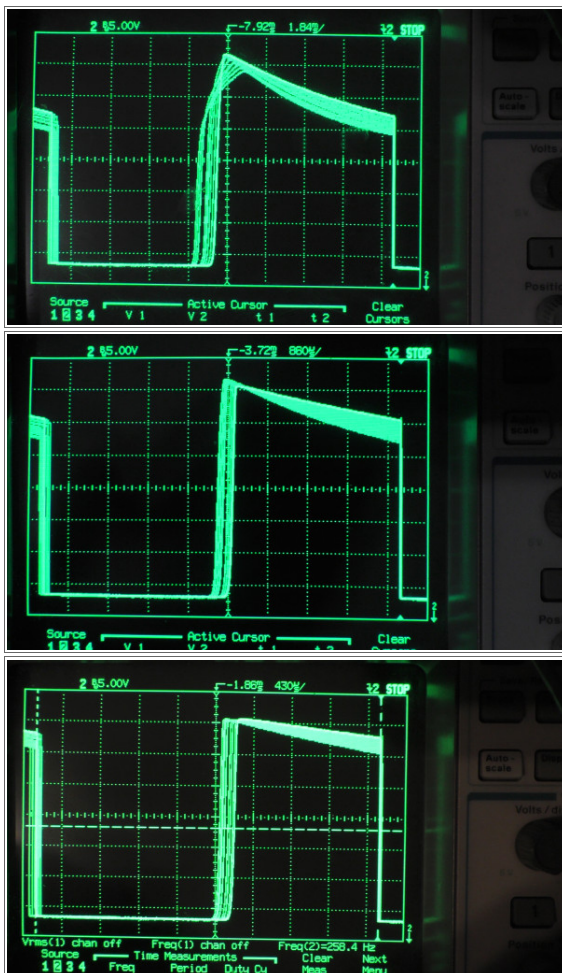
[toneburst](#)
Super Deluxe Wiggler

Posts: 1042
Joined: Fri May 02, 2014 2:41 pm
Location: London



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



the way I made these was by adjusting the vernier on the horizontal scale on my scope, and then doing a capture per note. you can see some jitter on the initial falling edge, as the vernier was too coarse to get an exact match. but, it gives a good idea of whats going on. you can see the falling slope on top of the square wave falls more slowly for higher frequencies, making it more "square" on top. at the same time, higher frequencies have a smaller pulse width. and then at very low frequencies, its all wonky.

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Re: 303 circuit modelling

by [toneburst](#) » Thu Jun 05, 2025 9:15 am

Great to see more progress!

Is the square wave made by running the saw through a comparator?

Re: 303 circuit modelling

by [guest](#) » Thu Jun 05, 2025 11:44 am

yes, but its a funky single transistor compator, and the comparison threshold is an RC circuit averaging the sawtooth, so at low frequencies this comparison point moves on the same time scale as the waveform, giving its funky shape.

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Re: 303 circuit modelling

by [guest](#) » Fri Jun 06, 2025 12:51 am

just for completeness sake, here are the lowest octave (achieved by turning the tune knob all the way the down), and the highest octave (tune knob all the way up, or pattern shifted up an octave):



“

[toneburst](#)
Super Deluxe Wiggler

Posts: 1042

Joined: Fri May 02, 2014 2:41 pm

Location: London

“



[guest](#)
Super Deluxe Wiggler

Posts: 13314

Joined: Mon Aug 19, 2013 11:49 am

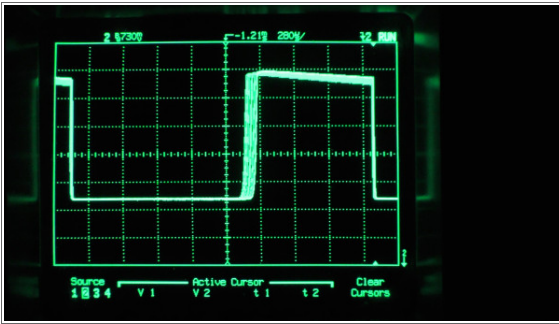
“



[guest](#)
Super Deluxe Wiggler

Posts: 13314

Joined: Mon Aug 19, 2013 11:49 am



I did the highest octave a little differently, where its not strictly notes as I matched the vernier on my scope to the frequency. this gives a more accurate look at the pwm, but less accurate with respect to note frequencies.

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Re: 303 circuit modelling

by [stepvhen](#) » Fri Jun 06, 2025 6:58 am

That low octave is definitely more of a suggestion than a square

(they/them)

I'm making Eurorack and 5U stuff: <https://www.gauche-electrosound.net>

new stuff: tiny pieces
harsh noise vaporwall
12 Sep 2025
[tiny pieces](#)

Re: 303 circuit modelling

by [guest](#) » Fri Jun 06, 2025 1:28 pm

definitely! its one of the things that makes the 303 sound. the square gets harsher and harsher the higher up in frequency you go.

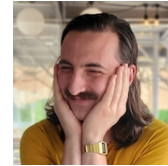
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“



stepvhen
Veteran Wiggler

Posts: 709
Joined: Sun Jun 07, 2020 10:41 am
Location: Indianapolis, USA
Contact: [@](#)

“



guest
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

127 posts

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